Loss Minimization in classification ⇒

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① Agenda ⇒

→ Recap

→ Distance b/w 2 lines | planes

→ Loss func^{n}

→ PERCEPTRON LEARNING RULE ⇒ 1st algo

Recap:

(a) Vectors: $\bar{x} \in \mathbb{R}^d$, $\bar{x} = \begin{bmatrix} x_1 \\ x_2 \\ \vdots \\ x_d \end{bmatrix}$ Column

(b) Norm: $\|\bar{x}\| = \sqrt{x_1^2 + x_2^2 + \cdots + x_d^2}$

(c) Inner Product (Dot): $\bar{x}, \bar{y} \in \mathbb{R}^d$

$\bar{x} \cdot \bar{y} = \bar{x}^T \bar{y} = \sum_{i=1}^{d} x_i y_i$

(d) Angle between vectors: $\cos\theta = \frac{\bar{x}^T \bar{y}}{\|\bar{x}\| \, \|\bar{y}\|}$

(e) Distance b/w origin and line:

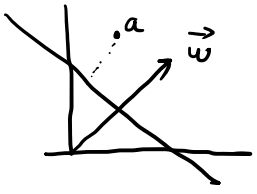
Line equation: $\bar{w}^T \bar{x} + w_0 = 0$ $\quad d = \frac{w_0}{\|\bar{w}\|}$

$\bar{w}$: weight vector; w_0: intercept

Distance ⇒ $d = \frac{w^T a + w_0}{\|w\|}$

$\Rightarrow \frac{w_1 a_1 + w_2 a_2 + w_0}{\|w\|}$

① =

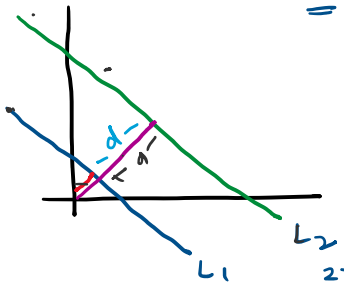


① from origin ⇒ $a = (0, 0)$

$\Rightarrow \frac{w_0}{\|w\|}$ ⇒ Distance of line from origin

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(*) Distance b/w 2 lines ⇒ lines are ∥ ⇒ parallel



L_1: $2x_1 + 4x_2 - 3 = 0$

L_2: $2x_1 + 4x_2 - 8 = 0$

$d = ?$

$$d = \frac{w_0^{green}}{\|w\|} - \frac{w_0^{Blue}}{\|w\|}$$

dist of L_2 from origin − Dist. of L_1 from origin

d = Dist. b/w 2 lines (*)

parallel lines ⇓

(*) slopes = Equal

$$\Rightarrow w = \begin{bmatrix} 2 \\ 4 \end{bmatrix}$$

same, only intercept ≠ of changing

(*) Shifting of lines

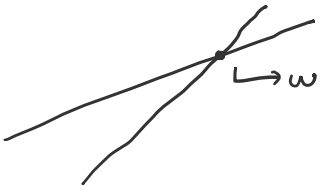
w is same for Both the lines

(*) Distance b/w 2 lines $= \frac{w_0^2}{\|w\|} - \frac{w_0^1}{\|w\|}$

(*) 2 lines ∥ Parallel ↳ distance = ?

0 ⇒

if lines are not ∥ ⇒ they are intersecting line



→ will meet at some point

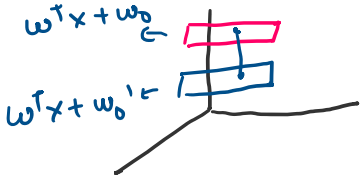
(*) Distance b/w 2 lines ⇒ only // lines ✓

⇒ extend this idea to any no. of dimension

easy → lines

$w^T x + w_0 = 0$

$w = \begin{bmatrix} w_1 \\ w_2 \end{bmatrix}$ → line



$w = \begin{bmatrix} w_1 \\ w_2 \\ w_3 \end{bmatrix}$ → plane

higher dim → hyperplane

⇒ $\left| \frac{w_0}{\|w\|} - \frac{w_0'}{\|w\|} \right|$ ⇒ exact same thing as line ✓

Dist. b/w 2 planes ⇒

Ques ⇒ $w^T x + 5 = 0$, $w^T x + 3 = 0$

distance

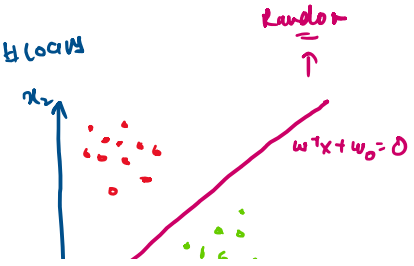
$w = [3, 2, 1]^T$

⇒ $d = \frac{5}{\sqrt{3^2+2^2+1^2}} - \frac{3}{\sqrt{3^2+2^2+1^2}}$

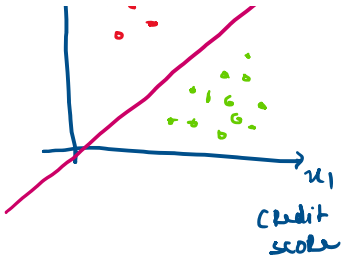
$= \frac{2}{\sqrt{14}} = 0.53$

(*) Loan Approval Example ⇒

→ need to find a seperator



→ need to find a seperator



To plot a line → what all things we need?

① w & w_0 ⇒ Best

$w = \begin{bmatrix} 2 \\ 4 \end{bmatrix}$ & $w_0 = -8$ ⇒ we can plot a line

New Applicant → $[400, 900]$ =

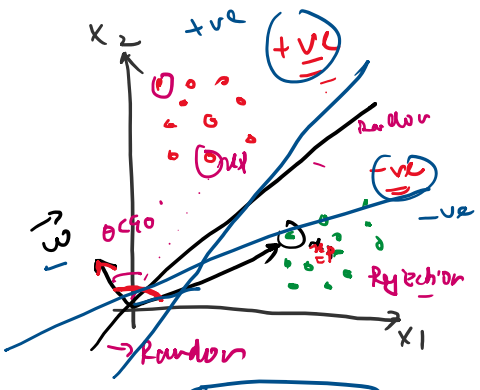
(*) which half the point lies in?

(*) sign of the distance → +ve → class 1 (Red)
→ -ve → class 0 (Green)

(*) Distance = $\vec{x_p} \cdot \vec{w} = \|x_1\| \|w\| \cos\theta$ (+ve, +ve, +ve or -ve, -ve)

(*) = -ve (-ve)

$\cos\theta$ = -ve ?



$\theta > 90° \Rightarrow \cos\theta$ = -ve

obtuse

(*) Best seperator =

(*) ① Best w & w_0 —
② which Half space the point lies in —
③ How confident are we with the req

X Our JOB is not to find Best w & w_0

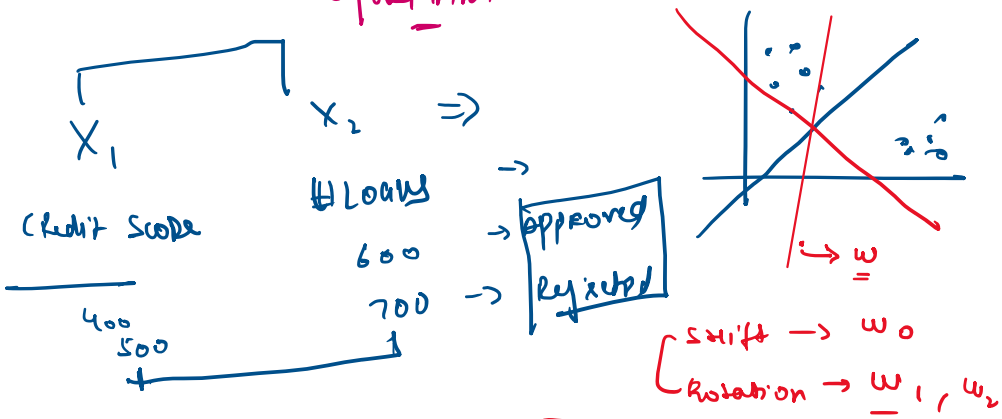
→ How do we know dir. of $\vec{w}$?

$\vec{w}$ X

values

Best w_0 & $\vec{w}$ = we can know where $\vec{w}$ is pointing

algorithm



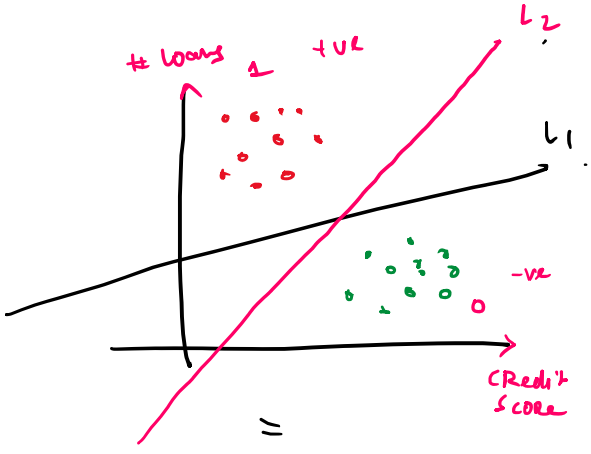
query → (3,2) $\sqrt{3^2+2^2}$ =

$w \rightarrow \frac{w}{\sqrt{w_1^2+w_2^2}}$ & w_0

① Algorithm

= * Hard spam

② Loss



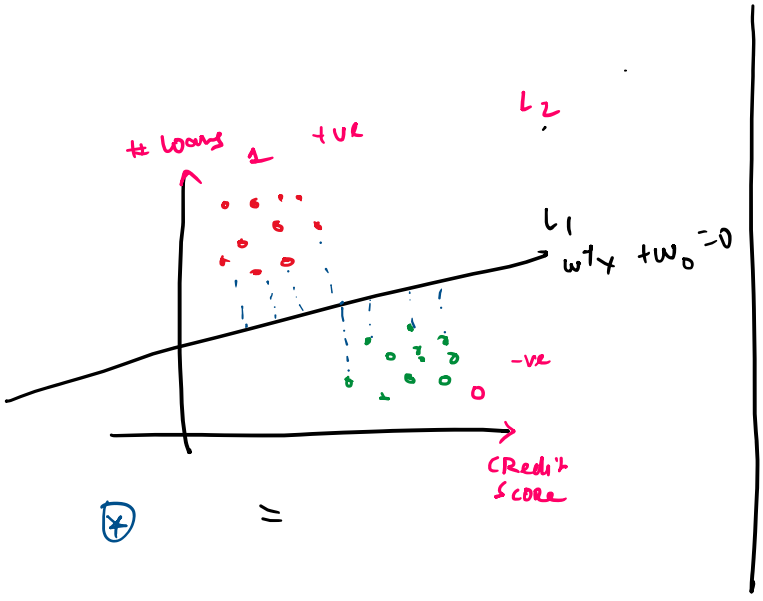
⇒ classifier front

Can separate these 2 classes = L_1 & L_2

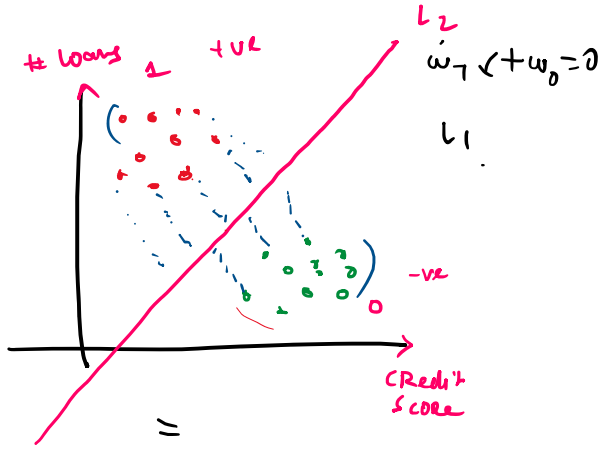
⊛ L_2 is better

⇒ Distance perspective

farther the points ⇒ more confident we are



⊛ =



=

⇒ in which case avg. distance from line would be greater

↓ maximum ⇒ Best separator

⊛ L2 separator

⇒ we want avg. distance maximized

⇒

⊛ How do i quantify L2 is better?

(*) How do i quantify L2 is better?

(*) $L_1 \rightarrow g(w_1, w_0) = 18$

$L_2 \rightarrow g(w_1, w_0) = 30$

func^n $\Rightarrow$ w_1, w_0

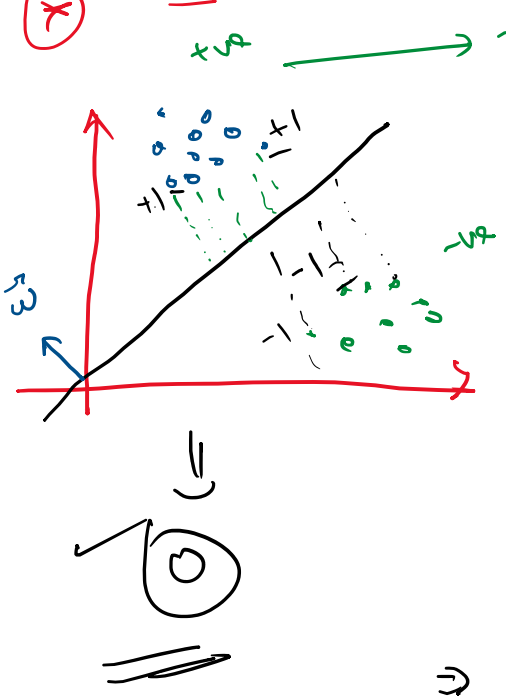
line

arg. distance

$\Rightarrow$

Task $\rightarrow$ func^n that can quantify how good or how bad a line is?

(*)



Total distance

$+1 + 1 - 1 - 1$

$= 0$ $\Rightarrow$ cancelling out each other

$\Rightarrow$ nullity effect

$\Rightarrow$ we want distances to be large

$\Rightarrow$ nullity effect

Solutions

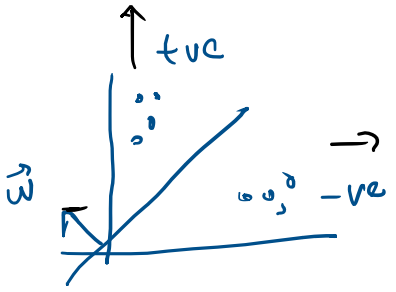
(1) ABS $\rightarrow$ $|-5| = 5$ PROBLEM $\rightarrow$ ABS optim

(2) Square $\rightarrow$ ✓

Squares ② Squares ⇒ ✓

+ve x +ve = +ve

$y_i = +1$



→ $y_i = -1$

-ve x -ve = +ve

③ multiply DP with corresponding y_i's

actual Labels

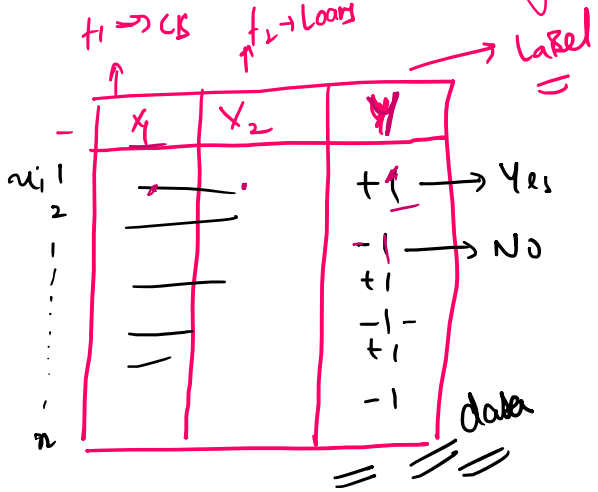
makes the distances +ve

⇒ find dist. of DP from Line & multiply with corresponding y_i & take avg.

⑥ Break ⇒ 10:24

$w^T x + w_0 = 0$

⇒ mathematically



distances Label

for any x_i → d_i

$$= \frac{w^T x^{(i)} + w_0}{\|w\|} \quad (*)$$

⇒ $x^{(i)} \cdot y^{(i)}$

$$= \left(\frac{w^T x^{(i)} + w_0}{\|w\|} \right) \cdot y_i$$

distance	Label	
+ve	+ve	= +ve
-ve	-ve	= +ve

$$\left(\frac{w \cdot x^i + w_0}{\|w\|}\right) \cdot y^i$$

↳ distance × (y^i) ↓ Actual Label

⇒ do this for all data points

iterate over all the data points

For all the data points

$$\Rightarrow \left[\sum_{i=1}^{n} \left(\frac{w^t x^i + w_0}{\|w\|} \cdot y^i\right)\right] \times \text{average}$$

= Total Distance from all points

we want to maximise

$$q = \underset{w, w_0}{\text{argmax}} \left(\frac{1}{n} \sum_{i=1}^{n} \frac{w^t x^i + w_0}{\|w\|} \cdot y(i)\right)$$

↳ optimization PROBLEM

↳ avg. dist. of all Data points from

↳ optimization PROBLEM

Data points from the line

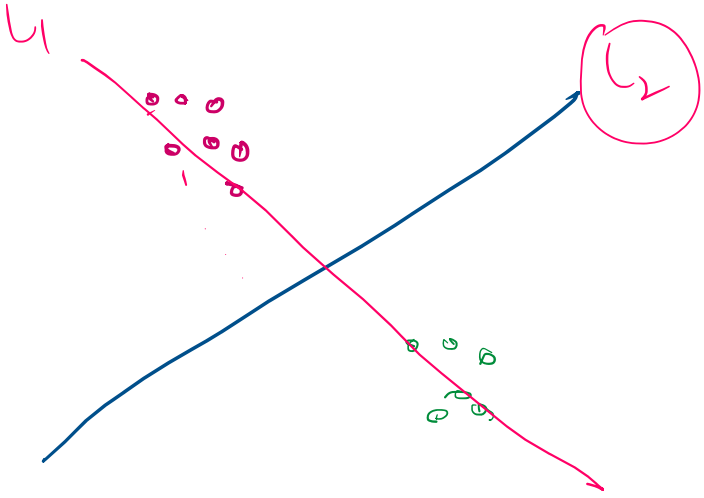
→ we need to find / optimize for Best w & w_0

Best Line

⇒ Best Line | Best seperator

⇒ (K) find such w & w_0 so that avg. dist. of points from line is maximized

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Best ⇒

L2

⇒ ALGORITHM

⇒ avg distance of points from line

⇒ avg distance of points from line
= gain function

(*) gain function = maximizing the gain function

↕

opposite

Loss function = = gain function

max / min.

↳ minimize

(*) PERCEPTRON ⇒ 30-40 min

X ⇒ ALGORITHM ⇒ weight & wo

gain func^n → Maximize

↳ profit, Reward

↳ profit, Reward,

Loss funⁿ ⇒ minimize

↳ error, cost, Loss

(+) maximize (gain) ⇒ minimize (−gain)

gain funⁿ ⇒

gain = 10 then Loss = −10

gain

10

5

1

Loss = −gain

−10

−5

−1

ML ⇒ Likel

Manhattan Distance →

① Manhattan city ⇒

Car → moving city

↑ ↓ → ←

Euclidean → shortest

(x_1, y_1) (x_2, y_2)

② $|x_1 - x_2| + |y_1 - y_2|$

2 → +ve

Grid path ↑ ↓ → ←

Euclidean → shortest distance